

KRISHNA N AGARAM

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EDUCATION

University of Illinois Urbana-Champaign, MS (Thesis track), Computer Science Urbana, IL
Advisor: Prof. Saurabh Gupta; research in **robotics** Aug. 2025 – May. 2027

- Coursework: Statistical RL Theory (A+), Advanced NLP (A), Deep Generative Models*, Computer Vision*

Indian Institute of Technology Bombay, B.Tech (with Honors), Computer Science Mumbai, India
GPA: **9.86/10**, ranked **3/194** overall; minors in **machine intelligence** and **mathematics** Nov. 2021 – May. 2025

- Selected coursework: RL, Robotics, Stat learning, Kernel methods, Formal methods for ML, Crypto, Advanced compilers, Discrete Math/Algo/Automata, Arch/Net/OS/DB/Compilers, Real/Complex/Fourier analysis, Linear/Abstract algebra

PREPRINTS

- *Quantum Advantage in Proof Systems without Entanglement*[†] Feb. 2026
K. Agaram, N. Spooner, Y. Zheng. Under review at **ICALP 2026**
- *Too Polite to Disagree: Understanding Sycophancy Propagation in Multi-Agent Systems*[†] Jan. 2026
K. Agaram*, V. Kasprova*, A. Parulekar*, A. Alrabah*, R. Garg, S. Jha. Under review at **COLM 2026**
- *Preparing arbitrary stabilizer states via disentangling and path-aware reinforcement learning*[†] Nov. 2025
K. Agaram*, S. Midha, V. Garg. **QIP 2025** and **ML4PS@NeurIPS 2025**; **oral** at the APS Global Physics Summit 2025
- *On the value and optima of the XOR-monogamy-of-entanglement game*[†] Jul. 2025
K. Agaram. Was coincidentally solved and put up by another group before we could publish; the draft is available [here](#)

RESEARCH EXPERIENCE

Sim2real for precise bimanual manipulation | Guide: Prof. Saurabh Gupta Aug. 2025 – present, UIUC

- Working on sim2real transfer for precise **bimanual** manipulation tasks focusing on generalization across objects
- Techniques unite ideas from abstract scene-object **representation** and RL **exploration** with few in-sim demonstrations

Sycophancy in multi-agent systems[†] | Guide: Prof. Dilek Hakkani-Tür Aug. 2025 – Dec. 2025, UIUC

- Quantitative evaluation of sycophantic behavior of a hierarchy of **cooperative multi-agent** systems across various tasks
- Designed novel pipeline using evaluated sycophancies to mitigate sycophantic behavior & improve accuracy of the system

Towards proving the Neural Feature Ansatz | Guide: Prof. Parthe Pandit Jun. 2025 – Sep. 2025, IIT Bombay

- Partial proof of the **Deep Neural Feature Ansatz** for the cosine kernel; this provably yields **fast neural feature learning**
- Proofs involve the use of neural tangent kernels, infinite matrix theory, some Fourier and functional analysis

Quantum Positional Proof Systems[†] | Guide: Prof. Nick Spooner Jun. 2024 – Sep. 2025, EPFL

- Formalized positional interactive proof systems; **characterized** complexity-theoretic expressivity in the classical setting
- Showed that quantum resources can improve expressivity using techniques from nonlocal games & **no-signaling theory**
- Proved that surprisingly, XOR soundness anti-concentrates on parallel-repetition of the monogamy-of-entanglement game[†]

Lower bounds on testing 3-colorability[†] | Guide: Prof. Akash Kumar Jan. 2024 – Apr. 2024, IIT Bombay

- Established **optimal** linear lower bound for one-sided 3-colorability testing on $(1/3 - \epsilon)$ -far vs colorable expander graphs
- Studied various **lower bound constructions** in an attempt to prove two-sided bound for ϵ -far vs colorable expanders

RL for quantum state preparation[†] | Guide: Prof. Vikas Garg Jun. 2023 – Sep. 2024, Aalto University

- Improved the quantum state preparation pipeline with an RL agent capable of constructing 9-qubit stabilizer states **zero-shot** ($\sim 4 \times 10^{16}$ states in total) while being **30%** more efficient than baselines; exploration solved via novel dense reward
- **Proved** that this efficiency is guaranteed for **at least 95%** of states despite seeing only **$10^{-8}\%$** of all states during training

OTHER PROJECTS

Mobile Robot Localization 🌐 | Guide: Prof. Leena Vachhani Oct. 2024 – Nov. 2024, final project: Robotics

- Wrote a ROS package for mobile robot navigation using **MoCap+trilateration** with Kalman filtering to offset sensor noise

Video Style Transfer 🌐 | Guide: Prof. Preethi Jyothi Aug. 2023 – Nov. 2023, final project: ML

- Implemented style transfer for videos, preserving style of moving objects across frames via occlusion-aware **optical flow**

READING PROJECTS

- ▶ **Probabilistic Proof Systems**: Examined the complexity-theoretic foundations of probabilistic proof systems, covering interactive proofs, Succinct ARGuments, PCPs and IOPs, from [Proofs, Arguments, and Zero-Knowledge](#)
- ▶ **Fast single-node PageRank**: Explored fast algorithms for single-node PageRank; wrote a unifying [survey](#) bringing out similarities between different existing algorithms; also implemented [this](#) very fast, [essentially optimal](#) algorithm
- ▶ **Analytic Combinatorics**: Studied the symbolic method arising from Pólya's enumeration theorem for combinatorial structures; applied it extensively to enumeration problems, exact and asymptotic, from [Analytic Combinatorics](#)

SCHOLASTIC ACHIEVEMENTS

- Selected to be a **Google DeepMind** pre-doctoral researcher (declined for MS at UIUC) Apr. 2025
- Received the **Institute Academic Prize** given to the **top 1%** of students for stellar academic record 2022, 2024
- Placed **1st, 2nd, 2nd** in India and **8th** (x3) internationally at the **Simon Marais Mathematics Competition** 2022, 2023, 2024
- Secured All India Rank **40** in **JEE Advanced** and 122 in JEE Main among 140K+ and 1M+ aspirants respectively 2021
- Among the **top 35** students in India (x2) invited to the **International Mathematics Olympiad Training Camp** 2020, 2021
- Qualified for the International Olympiad on Astronomy and Astrophysics Team Selection Camp (IOAA OCSC) 2020
- Conferred with the prestigious **KVPY** (All India Rank 23) and **NTSE** science scholarships 2019, 2020

Teaching and Mentorships

Teaching Assistantships

- [UIUC] CS 374, Algorithms and Models of Computation**
Held problem-solving sessions twice each week, set and graded tests Fall 2025, Spring 2026
- [IITB] CS 785, Theoretical CS Toolkit** (Graduate)
Taught bi-weekly tutorial sessions; set and graded problem sheets and exams. Spring 2025
- [IITB] CS 215, Data Analysis.** Set and graded quizzes and exams for a class of **200** Fall 2024
- [IITB] CS 213, Data Structures and Algorithms**
Held bi-weekly problem-solving sessions, created programming lab assignments, set and graded tests Fall 2023
- [IITB] MA 106, Linear Algebra.** Conducted **weekly** tutorial sessions for **40+** freshmen Spring 2023, Spring 2024

High School Tutoring

- Taught Physics & Chemistry **one-on-one** as part of the **National Service Scheme** (NSS) Jun. 2024 - Aug. 2024
- Worked with the **Vizuara** team in developing short animated videos **motivating math concepts** in early high-school Math for use in schools, using the Python animation library **Manim** Oct. 2022 - Dec. 2022
- Delivered **lectures** covering topics in [Symbolic Combinatorics](#), [Barycentric Coordinates](#), [Generating Functions](#) and [Projective Geometry](#) to interested high-school students as part of the **Online Math Club** Nov. 2021 - Dec. 2022

TECHNICAL SKILLS

Languages	Python, C/C++, Java, x86 assembly, SQL, bash, MATLAB, Javascript, $\LaTeX$
Frameworks	PyTorch, ROS, NVIDIA IsaacGym/Lab, Drake, MuJoCo, OpenAI Gym, NumPy, Pandas, <i>et al.</i> , IBM Qiskit

[†] Preprint available upon request. * Ongoing.